MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4DSCMOS200 Wireless Communication Technology

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	Wireless communication uses which type of waves for transmission? a) Mechanical waves b) Sound waves c) Electromagnetic waves d) Seismic waves	U	1
2	The frequency range of the UHF band is: a) 3–30 kHz b) 30–300 kHz c) 300 MHz–3 GHz d) 3–30 GHz	U	1
3	The main purpose of modulation is to: a) Increase noise b) Reduce bandwidth c) Enable long-distance transmission d) Remove the carrier	U	1
4	Which block is responsible for converting information into a suitable signal for transmission? a) Receiver b) Channel c) Transmitter d) Destination	U	2
5	CDMA allows multiple users by using: a) Different frequencies b) Different time slots c) Different spreading codes d) Different antennas	An	2
6	Which wireless technology is mainly used for short-range, low-power communication? a) Wi-Fi b) Bluetooth c) GPS d) GSM	U	2
7	Amateur (HAM) Radio is primarily used for: a) Commercial broadcasting b) Paid communication services c) Emergency and non-commercial communication d) Television transmission	U	3
8	Which of the following is a commonly used SDR hardware platform? a) Arduino Uno b) Raspberry Pi c) RTL-SDR d) PIC Microcontroller	U	3

9	The basic concept of SDR is to replace hardware using: a) Filters b) Antennas c) Software d) Power amplifiers	U	3
10	GNU Radio is mainly used for: a) Circuit simulation b) Network routing c) Software Defined Radio implementation d) Database management	U	3

1 * 10 = 10

Part B

Short Answer Questions
Answer any **4** Questions
Each question carries 5 marks

11	Compare ASK, FSK, and PSK digital modulation techniques.	An	1
12	Explain the basic building blocks of a communication system.	U	2
13	Explain Bluetooth technology and its applications.	U	2
14	Write short notes on CDMA and its advantages.	U	2
15	Describe the need for Software Defined Radio (SDR).	U	3
16	Write short notes on Amateur (HAM) Radio.	U	3

4 * 5 = 20

Part C

Essay Questions
Answer any **2** Questions
Each question carries 10 marks

17	Explain in detail the concept of wireless communication, signals, and systems.	U	1
18	With neat block diagrams, explain Amplitude Modulation (AM)	U	1
19	Explain the 2G (GSM) cellular network architecture with a block diagram.	An	2
20	Describe GNU Radio, its features, and explain how SDR hardware is connected using GNU Radio.	An	3

2 * 10 = 20